BLOOD BANK MANAGEMENT MODULE

Business Analyst Case Study — Healthcare Information Systems | Portfolio Project

ROLE	INDUSTRY	ENGAGEMENT	DELIVERABLE
Business Analyst	Healthcare / Hospital IT	Self-directed portfolio project	Business Requirements Document

THE CHALLENGE

Hospital blood banks coordinate donor registration, donation, blood-unit lifecycle management, inventory and patient requests raised from IP/OP/ER/Day Care/OT — all dependent on accurate, timely data. Without a structured digital workflow, teams relied on fragmented manual records, making it difficult to see current stock, track request status, identify near-expiry units, or prove an auditable end-to-end history of a unit.

MY APPROACH

- Mapped 8 stakeholder groups (Administrator, Technician, Clinician, Nurse, Patient, System Admin, Management, QA) and their distinct needs.
- Elicited 25+ functional requirements across authentication, donor, donation, unit, inventory, request, issue, audit and reporting, each prioritised as Critical, High or Medium.
- Modelled the As-Is process against a redesigned 9-step To-Be workflow built around explicit unit and request statuses.
- Defined 15 business rules and a 7-entity data model (Donor, Donation, Blood Unit, Patient, Request, Allocation, Issue, Audit Event).
- Built traceability from business objectives through to functional requirements and use cases, and authored 9 use cases as the basis for UAT and acceptance criteria.

THE SOLUTION

A workflow-driven To-Be process: technicians register or search donors and record donations, generating traceable unit records; authorised staff progress units through configured lifecycle statuses into available inventory; clinicians raise requests that move through review, allocation and issue, with every step captured in an audit trail. Core capabilities: role-based access, eligibility validation before issue, expiry monitoring, request status tracking (Draft → Pending → Approved/Rejected → Fulfilled/Cancelled), and operational dashboards and reports.

THE RESULT

- Faster request handling through a visible, status-driven request lifecycle from creation to fulfilment.
- Fewer errors via mandatory-field and business-rule validation applied at the point of entry.
- Stronger traceability, with every critical transaction attributable to a user and timestamp.
- Better stock control through status-derived inventory counts and expiry/low-stock alerting.
- Reduced manual reporting effort via standard dashboards and exportable operational reports.

MEASURABLE OUTCOMES (TARGETS)

8 → 5	100%	0
Reduction in manual request-processing steps (target)	Traceability achieved for priority/critical requirements (target)	Expired or incompatible unit issues, via automated validation (target)

SKILLS DEMONSTRATED

Requirements Elicitation	Stakeholder Analysis	Process Modelling	Business Rules Definition
Data Modelling	Use Case Writing	Requirements Traceability	UAT & Acceptance Criteria
Java / Spring Boot	Angular	PostgreSQL	Redmine